

A *mapping* is a data structure that connects two or more pieces of information, commonly called a *key-value pair*. When you ask for a key, you get its associated value. One mapping structure can store many key-value pairs.

Mappings don't just store data. They allow you to establish relationships between the data. For example, you could pair each username with a password, each country with its capital, and each book with its author.

Mappings are not limited to single pieces of data: you could use a single piece of data as a key and use a sequence as the value. For example, you could pair a username with a sequence of other users to define a relationship between one user and other users. This is the core of how many social networks are implemented.

In Python, the *dictionary* is the main mapping type. Make sure you understand dictionaries, because many advanced Python data structures are represented by dictionaries.